Lecture 34 (*May 17*)

Reading: Section 10.5
Viewing: none

We will now study Fourier series, which is a very useful application of linear algebra in analysis, and from there in applied mathematics and engineering. But here's the twist: Fourier analysis is not concerned with finite-dimensional vector spaces, but with infinite-dimensional ones.

Definition 30. *Let V be the vector space of differentiable functions $f : \mathbb{R} \to \mathbb{R}$ with period 2π, i.e.:*

$$f(x + 2\pi) = f(x) \qquad \forall x \in \mathbb{R} \tag{321}$$

Such functions are called ***periodic****.*

The fact that we only look at periodic functions might seem like a restriction, but it's suitable for many applications (think signals, waves). Also, the fact that the period is 2π rather than any other number is not a meaningful restriction, but more of a notational convention. The reason for this is that if $f(x)$ is a function with period L, then $f(\frac{2\pi x}{L})$ is a function with period 2π.

The fact that the set of f's as in (321) is a vector space is easy to see, basically because any linear combination of differentiable periodic functions is differentiable and periodic (this entails the fact that all of these functions have the same period, but we have already assumed that this is the case). To convince you that it is not a finite-dimensional vector space, consider the functions:

$$1, \cos x, \cos 2x, \cos 3x, \ldots, \sin x, \sin 2x, \sin 3x, \ldots \tag{322}$$

We claim that all of the infinitely many functions in (322) are linearly independent: there is no <u>linear</u> combination of these functions which is equal to the zero function [5]. Therefore, the space V is an entirely new beast from the vector spaces $\mathbb{R}^n$ that we have studied in most of our course. However, it does share a lot of features with finite-dimensional vector spaces, such as the existence of a dot product like operation, namely the **inner product** of functions:

$$(f, g) = \int_{-\pi}^{\pi} f(x)g(x)dx \tag{323}$$

[6] In other words, the inner product takes two functions and produces a number, just like the dot product took two vectors and produced a number. And just like the length of a vector is its dot product with itself, we have the **norm** of a function:

$$||f||^2 = (f, f) = \int_{-\pi}^{\pi} f(x)^2 dx \tag{324}$$

which is a positive real number (unless $f = 0$, in which case the norm is 0). Here's one important fact: the functions (322) are all mutually orthogonal with respect to the inner product (323):

$$(\sin kx, \sin lx) = 0, \quad \text{unless } k = l, \text{ in which case } (\sin kx, \sin kx) = \pi \tag{325}$$

$$(\cos kx, \cos lx) = 0, \quad \text{unless } k = l, \text{ in which case } (\cos kx, \cos kx) = \pi \tag{326}$$

$$(\sin kx, \cos lx) = 0, \quad \text{for all } k, l \tag{327}$$

[5] Note the emphasis on the word "linear", since there are non-linear combinations of the above functions which are equal to the zero function, for example $(\sin x)^2 + (\cos x)^2 - 1 = 0$. However, when studying vector spaces, all we are concerned with are linear combinations

[6] The fact that the integral goes from $-\pi$ to π is not important. You could have taken it over any interval of length 2π and the value of the integral would have been unchanged, due to the periodicity of the functions f and g

The formulas above also hold if k or l are 0 (except that $\sin 0 = 0$ and $\cos 0 = 1$ imply that $(0,0) = 0$ and $(1,1) = 2\pi$). So (322) is a collection of linearly independent, mutually orthogonal elements of the vector space V. If this were a finite-dimensional vector space, we could hope that the collection (322) is also a basis. Well, something pretty close to this actually happens:

Fact 23. *Any function $f \in V$ can be written as a linear combination of the functions* (322)*, but the linear combination has to be infinite. In other words, we can write any function $f \in V$ as:*

$$f(x) = a_0 + a_1 \cos x + a_2 \cos 2x + \cdots + b_1 \sin x + b_2 \sin 2x + \ldots \tag{328}$$

The numbers $a_0, a_1, a_2, \ldots, b_1, b_2, \ldots$ are called the ***Fourier coefficients*** *of $f(x)$, and the sum* (328) *is called the* ***Fourier series*** *of $f(x)$. It is a theorem that the Fourier series converges.*

In signal processing (and music) the functions $\cos kx, \sin kx$ are called **harmonics**, so the Fourier series is basically just a way to decompose an arbitrary periodic signal $f(x)$ into harmonics. As you would expect, in real-life examples the coefficients a_n, b_n tend to 0 as $n \to \infty$, so only the first few harmonics of a signal dominate it.

Linear algebra also gives us a very practical way to compute the Fourier coefficients of any function. Because the functions (322) are orthogonal, you can compute these coefficients by computing the inner product of f with any one of these functions:

$$(f, 1) = a_0(1,1) \qquad \Rightarrow \qquad a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx \tag{329}$$

$$(f, \cos kx) = a_k(\cos kx, \cos kx) \qquad \Rightarrow \qquad a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)(\cos kx)dx \tag{330}$$

$$(f, \sin kx) = b_k(\sin kx, \sin kx) \qquad \Rightarrow \qquad b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)(\sin kx)dx \tag{331}$$

Formulas (329), (330), (331) show that if you have a machine that computes various integrals of f (and there are many ways for getting approximate answers) this immediately gives a recipe for computing the Fourier series of f. Here's an example of this setup. Consider the function:

$$f(x) = x \quad \text{for all } x \in [-\pi, \pi] \tag{332}$$

then extended to all real numbers x so that it has period 2π (note that this function has a discontinuity at $x = \pi$; all the facts in this class still hold in this slightly more general setup). To compute its Fourier series, we need to compute the following integrals:

$$(f, \cos kx) = \int_{-\pi}^{\pi} x(\cos kx)dx = 0$$
$$(f, \sin kx) = \int_{-\pi}^{\pi} x(\sin kx)dx = (-1)^{k-1}\frac{2\pi}{k}$$

which can be done e.g. by integration by parts. Therefore (329)–(331) imply the Fourier series:

$$f(x) = \frac{2\sin x}{1} - \frac{2\sin 2x}{2} + \frac{2\sin 3x}{3} - \frac{2\sin 4x}{4} + \ldots \tag{333}$$

The fact that the Fourier series only contains sines in this example should not be surprising; after all, the particular function $f(x)$ in (332) is odd. But then let us compute the norm squared of the function (332). On one hand, we have:

$$(f,f) = \int_{-\pi}^{\pi} x^2 dx = \frac{2\pi^3}{3}$$

On the other hand, the norm squared of the right-hand side of (333) is easy to compute, because the functions $\sin kx$ have norm π and pairwise zero inner product:

$$(f,f) = \frac{4\pi}{1^2} + \frac{4\pi}{2^2} + \frac{4\pi}{3^2} + \frac{4\pi}{4^2} + \dots$$

By equating the two expressions above, we arrive at the interesting identity:

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$$

This (and others like it) is a neat identity involving π, which gives an easy way to approximate the powers of π. Based on the formula above, you could even approximate π using pen and paper (assuming you remember the algorithm for taking square roots by hand).

Let's now discuss a variant of Fourier series, wherein we replace the vector space V from Definition 30 by the vector space of differentiable functions with complex (instead of real) values:

$$f : \mathbb{R} \to \mathbb{C}$$

with period 2π. Because complex numbers are of the form $a + bi$ where a, b are real numbers, such complex-valued functions are of the form $a(x) + b(x)i$ where $a(x), b(x)$ are real-valued functions. Much of the theory concerning Fourier series carries through in the present setup, but in a sense it's simpler and more elegant. This is because instead of working with the particular real-valued functions (322), you can work with the complex-valued functions:

$$e^{ikx} \quad \text{for all integers } k \tag{334}$$

In fact, the functions in (322) are merely the real/imaginary parts of the functions in (334). In the complex-valued setting, the definition of inner product and norm from (323) should be changed to:

$$(f,g) = \int_{-\pi}^{\pi} f(x)\overline{g(x)}dx \tag{335}$$

where $\overline{z}$ denotes the complex conjugate of the complex number z. The reason for this is that you want the norm of any function to be a non-negative real number:

$$||f||^2 = \int_{-\pi}^{\pi} |f(x)|^2 dx \geq 0 \tag{336}$$

and you would not have been able to achieve this without the conjugate in (335) (what I'm saying is that if z is an arbitrary complex number, z^2 is also a complex number, but $z\overline{z} = |z|^2$ is a non-negative real number). It is then easy to compute the inner products of the functions (334):

$$(e^{ikx}, e^{ilx}) = \int_{-\pi}^{\pi} e^{ikx}\overline{e^{ilx}}dx = \int_{-\pi}^{\pi} e^{i(k-l)x}dx = \begin{cases} \frac{e^{i(k-l)x}}{i(k-l)}\Big|_{-\pi}^{\pi} = 0 & \text{if } k \neq l \\ 2\pi & \text{if } k = l \end{cases}$$

So the functions (334) are pairwise orthogonal. The complex-valued analogue of Fact 23 is that any differentiable complex-valued periodic function has a Fourier series of the form:

$$f(x) = \sum_{k \in \mathbb{Z}} c_k e^{ikx} \tag{337}$$

where the Fourier coefficients are given by:

$$c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx \tag{338}$$

The advantage of complex-valued Fourier series over real-valued ones is clear: the Fourier series (337) is a sum of only one type of function, as opposed from two types of functions in (328). Moreover, you only need to remember the single formula (338) for the Fourier coefficients in the complex-valued case, instead of formulas (329), (330) and (331) in the real-valued case. Finally, since any real-valued function can also be thought of as a complex valued function, it still has a Fourier series of the form (337) (it's just that you will have $\overline{c_k} = c_{-k}$, in order for the imaginary parts of various complex exponentials to cancel each other out).

Lecture 35 (*May 19*)

Review and more examples on the previous class.